

VOLUME Theorems

Theorems of the Project

T1-T10: statements, lemmas, proofs and honest labels

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About this volume

This volume collects the ten statements on which the whole series rests. Each carries a status label: proven — a complete proof plus a machine verification exists; measured — a statement about a finite computational experiment with controls; conjecture — a working hypothesis of the series. We deliberately do not inflate the status: the difference between a proof and a measurement is part of the method.

The numbering T1–T10 is shared by the READMEs, the monographs C1–C8, the Hilbert–Pólya scorecard and the Independent Verification Protocol. The proofs are given in a volume sufficient for checking without external sources; every constant is derived inside the series.

T1–T3: the frame and the counting

T1 (Γ -closures) — proven.

$|\Gamma(\frac{1}{2}+iu)|^2 = \pi/\cosh \pi u$; $|\Gamma(iu)|^2 = \pi/(u \cdot \sinh \pi u)$; $|\Gamma(\frac{1}{4}+iu)| \cdot |\Gamma(\frac{3}{4}+iu)| = \pi\sqrt{2}/\sqrt{\cosh 2\pi u}$; the single modulus on the line admits no elementary closure (K1, refuted).

Proof. Euler reflection, Legendre duplication; the counterexample to the candidate form is 0.765 at $u = 1$. Verification at $dps = 40$: 1.3–2.6e-39.

T2 (reality of Z_{id}) — proven.

$\text{Im } Z_{id}(t) = 0$ on the critical line; parity holds to 9.9e-29.

Proof. $\xi(\frac{1}{2}+it) \in \mathbb{R}$ (the symmetry $\xi(s) = \xi(1-s)$), $\theta(t)$ odd. Verification at $dps = 30$.

T3 (pole calibration) — proven.

$N(T) = \Delta \arg \xi/2\pi$ with no additive +1 when the pole $s = 1$ is bypassed from above; a single leaf condition loses turns (2π aliasing).

Proof. The residue of ξ'/ξ at $s = 1$ is +1; a clockwise half-passage contributes $-\pi$ to $\Delta \arg$. Practical verification: anchors 0/3/10/29/79/649 and the arithmetic bridge 2.6e-26.

T4–T6: arithmetic and the trace

T4 (co-location) — proven.

$(\Phi_p, \Phi_q) \equiv \Phi_{pq}$ with $\Phi_p = \frac{1}{2} \ln p$; $\max |\Delta E| = 1.35e-14$.

Proof. Additivity of the logarithm; the lattice realisation preserves the identity at machine level.

T5 (resolution kernel) — measured.

The composite-node response $\delta(d)$: $0 \rightarrow 3.2e-3$ ($d = 0.5$) $\rightarrow 1.5e-2$ ($d = 2$) $\rightarrow$ saturation $\sim 2e-2$ at $d \approx 4-6$; multiplicativity is localised in 2–3 cells; first-order additivity fails ($\sim 1e-2$).

Proof. Direct measurement on the grid $d \in [0, 6]$; honesty label: depends on the flow conventions of C6.

T6 (trace formula for H_{ξ}) — proven and verified.

$Q(a) = \zeta'/\zeta(1/2+a) + 1/(1/2+a) + 1/(a-1/2) - 1/2 \ln \pi + 1/2 \psi(1/4+a/2) - B - C_0$ with $B = 1/2 \ln(4\pi) - 1 - \gamma/2 = -C_0$; $K(x) = \text{Tr} e^{-xH_{\xi}} = (1/\pi) \int_0^\infty Q(a) \sin(ax) da$; for $a > 1/2$ the ζ -series contributes the weight $\Lambda(n)/\sqrt{n}$.

Proof. Hadamard product plus $2a/(a^2+\gamma^2) = 2 \int e^{-\gamma t} \sin(at) dt$. Verification: $|\Delta Q| \leq 1.7e-12$ (M0), $|\Delta K| \leq 1.2e-5$ (M1).

T7–T10: spectrum and symmetry**T7 (universality demarcation) — measured.**

87 gates (52 protocol + 35 extended) lie in the GUE class ($\langle r \rangle = 0.498\text{--}0.612$) and carry no Λ -weights: $r_{H7} \leq 0.226$, $|r_{H7}| \leq 0.043$ over the new handles, r_{EF} inside the null clouds.

Proof. Campaigns C3–C5 with positive controls PC1/PC2 ($r = 1.0$) and PC0 ($r = 0.9998$).

T8 (64→4 reduction) — proven.

The E2 spin structures collapse to 4 Hamiltonians by the pairs $(\varphi x, \varphi y) \bmod 2\pi$; isospectrality within classes $1.1e-14$.

Proof. The phases enter through two sums of three 1-forms $\bmod 2\pi \rightarrow 2 \times 2 = 4$ classes; verified against the E1/E2 tables.

T9 (zero modes at $\alpha = 1/3$) — measured.

On frustrated tori $L \equiv 0 \pmod{4}$ the clean lattice at $\alpha = 1/3$ has exact zero modes: 18 at $L = 12$, 42 at $L = 24$.

Proof. Direct diagonalisation with fixed multiplicities; a number-theoretic explanation (the Dirac index) is an open question of the series.

T10 (statistics ladder) — measured.

Poisson/GOE/GUE are governed by symmetry (frustration / real disorder / complex vortices); GSE is structurally unreachable ($T^2 = +1$).

Proof. An 11-channel ladder, classification by $\langle r \rangle + KS$; the GSE limit is architectural.

Summary table and consequences

Of the ten statements, four are fully proven (T1, T2, T4, T6), one is proven and verified to a numerical floor (T6), four are measured with controls (T5, T7, T9, T10), and one is proven at the level of phase structure (T8). None carries a status higher than its content — and it is exactly this discipline that lets the series speak of the H2 gap of 9–10 orders as a measured fact rather than an estimate.

The main consequences for the Hilbert–Pólya programme are collected in the scorecard: the resonator is built and tuned (H1+H3+H4), the zeros are encoded rather than generated (H2+H5), and the prime weight is verified only for the zero operator (T6). A hypothetical HP-operator, if it exists, is not a local deformation of the lattice family — the single bridge between all ten theorems and the three projections of the gap.

Table 1. *Statuses of the theorems*

T	Content (brief)	Status
T1	Γ -closures: 3 forms + a refutation	proven
T2	$\text{Im } Z_{\text{id}} = 0$	proven
T3	pole calibration of the argument principle	proven
T4	co-location $(\Phi_p, \Phi_q) \equiv \Phi_{pq}$	proven
T5	resolution kernel $\delta(d)$, $\xi_{\text{mult}} \approx 2-3$	measured
T6	trace formula $\text{Tr } e^{-xH_{\xi}}$	proven+verified
T7	universality demarcation (87 gates)	measured
T8	$64 \rightarrow 4$ reduction	proven
T9	zero modes at $\alpha = 1/3$	measured
T10	Poisson/GOE/GUE ladder	measured